

ROMAL RAMADHAN

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Research Interests

Research Interests: CO₂ Storage, H₂ Storage, Geothermal Energy, Critical Minerals, Geomechanics

Primary Fields: Reservoir Simulation, Coupled Subsurface Flow–Geomechanics, Fault Leakage, Caprock Integrity, Wellbore Stability, Containment Risk, and Monitoring Design and Uncertainty Analysis

Secondary Fields: Subsurface Containment Risk Assessment, Geologic Energy Storage Design & Monitoring and Data Integration

Skills

Technical Skills: CMG, Petrel, Eclipse, MBAL, Pressure Transient Analysis (PTA)

Data Analysis: Python, MATLAB, OriginLab, and L^AT_EX

Education

Ph.D. in Geoenergy and Geoenvironmental Engineering, Jackson School of Geosciences, The University of Texas at Austin, USA Aug 2024–2028 (expected)

Gulf Coast Carbon Center (GCCC), Bureau of Economic Geology (BEG)

Co-Advisors: Dr. Seyyed A. Hosseini; Dr. Larry W. Lake

Relevant Coursework: Reservoir Simulation, Advanced Reservoir Engineering, Subsurface Energy Storage, Enhanced Oil Recovery, Applications of Data Analysis, Visualization in Geosciences, Python and Machine Learning in Geosciences

M.Sc. in Petroleum Engineering, Chulalongkorn University, Thailand

Jan 2021–Jan 2023

B.Sc. in Petroleum Engineering, Universitas Islam Riau, Indonesia

Sep 2014–Jun 2018

Work Experience

Graduate Research Assistant, Gulf Coast Carbon Center (GCCC), Bureau of Economic Geology (BEG), UT Austin

Aug 2024–Present

- **Project 1:** *Boundary Condition Optimization for Efficient Compliant CCS Reservoir Simulation*
 - Developed storage-conserving boundary-modifier workflows for truncated-domain reservoir models, improving pressure-footprint and AoR fidelity while retaining high-throughput simulation speed.
 - Benchmarked reduced-domain models against full-domain reference simulations and quantified accuracy–efficiency tradeoffs to support uncertainty/sensitivity campaigns.
 - Translated modeling outcomes into regulator-facing deliverables (AoR defensibility, monitoring-zone implications) with reproducible model configurations and documentation.
- **Project 2:** *Fault Integrity and Geomechanical Risk Assessment for CCS Operations*
 - Built a fault-integrity screening workflow coupling flow simulation outputs with geomechanical failure metrics (effective stress evolution, stress rotation, slip tendency).
 - Ran scenario stress-tests across injection controls and fault-property uncertainty to define failure envelopes and safe operating margins.
 - Connected slip outcomes to consequence metrics (e.g., cross-confining-layer brine flux, AoR sensitivity) to inform monitoring priorities and mitigation options.
- **Project 3:** *Linking Poroelastic Theory to Field-Scale Monitoring of the Noordbergum Effect in CCS*
 - Developed a layered poroelastic interpretation framework for early-time pressure transients to separate mechanical-coupling effects from leakage-like signatures.
 - Designed paired numerical experiments (diffusion-only vs fully coupled flow–deformation) to quantify inference bias and identify when coupled interpretation is required.

- Converted transient diagnostics into monitoring design guidance (sensor placement/stratigraphic targeting and operational decision triggers).

Reservoir Engineer (Research), Chiang Mai Carbon Capture and Storage Frontier
Research Centre, Chiang Mai University

Jan 2023–Jun 2024

- Led multi-disciplinary modeling projects on CO₂ storage in deep saline aquifers, quantifying trapping (residual, solubility, mineral) and assessing caprock integrity via coupled geomechanical and geochemical simulations.
- Optimized CO₂ plume monitoring strategies by integrating spatiotemporal simulation with risk assessment, supporting publications on subsurface containment dynamics.
- Advanced simulation of pressure evolution and geochemical interactions to inform safe long-term injection and reservoir performance in complex geology.

Selected Peer-Reviewed Publications

1. **Ramadhan, R.** & Mon, M. T. (2026). *Underused Geothermal Potential in Southeast Asia's Net-zero Transition*. Geoscience Frontiers
2. **Ramadhan, R.** & Hosseini, S. A. (2026). *Gradual modifiers for storage-conserving truncated porous media models: accuracy and efficiency in CCS flow simulation*. Advances in Water Resources
3. **Ramadhan, R.**, Abdurrahman, M., Arsad, A., et al. (2025). *Reservoir heterogeneity and its role in long-term CO₂ storage performance: A case study of Air Benakat formation*. Deep Resources Engineering.
4. Tangparitkul, S., Akamine, T., **Ramadhan, R.**, et al. (2025). *CO₂ storage infrastructure and cost estimation for BECCS in northern Thailand*. Carbon Capture Science & Technology.
5. **Ramadhan, R.**, Tapanya, C., Thakheru, A., Leelasukseree, C., & Tangparitkul, S. (2024). *CO₂ trapping dynamics in tight sandstone: Insights into trapping mechanisms in Mae Moh's reservoir*. Journal of Environmental Management.
6. **Ramadhan, R.**, Mon, M. T., Tangparitkul, S., Tansuchat, R., & Agustin, D. A. (2024). *Carbon capture, utilization, and storage in Indonesia: Storage capacity, status, economics, and policy*. Energy Geosciences.
7. **Ramadhan, R.**, Promneewat, K., Thanasaksukthawee, V., et al. (2024). *Geomechanics contribution to CO₂ storage containment and trapping in tight sandstone complexes: Mae Moh Basin*. Science of the Total Environment.
8. **Ramadhan, R.**, Abdurrahman, M., Bissen, R., Maneeintr. K. (2023). *Numerical simulation of potential site for CO₂ sequestration in a depleted oil reservoir in northern Thailand*. Energy Report.

Presentations

Poster: "Sealing or leaking? Evaluating fault integrity under CO ₂ storage operations" Jackson School Annual Research Symposium, Austin, TX	2026
Oral: "Efficient Boundary Modeling for CO ₂ Storage in Saline Aquifers" GSA Connect, San Antonio, TX	2025
Poster: "Reduced-grid models with boundary-condition accuracy for subsurface energy and waste storage" Jackson School Annual Research Symposium, Austin, TX	2025
Oral: "Numerical simulation of potential site for CO ₂ sequestration in a depleted oil reservoir in northern Thailand" ICREC, Paris, France	2022
Oral: "Correlation comparison for prediction of density and viscosity of aqueous 2-(ethylamino)ethanol (EAE) for carbon capture" AUA Academic Conference, Kuala Lumpur, Malaysia	2022

Awards, Fellowships and Activities

CCUS Conference Week 2025 — Whole Value Chain CCUS	2025
SMART-CDR Competition (Carbon Dioxide Removal) — Semi-finalist	2024
Chevron Scholarship for Graduate Student ~USD 2000	2023
Chulalongkorn University's Graduate Scholarship Programme	2021-2023